

CS 663: Digital Image Processing

Assignment 1: Problem 2(a) and 2(b) Report

Pushpendra Uikey

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1 Thresholding

1.1 2(a) Manual Global Thresholding

Implementation

Global thresholding segments an image by comparing every pixel intensity against a single, user-defined scalar T . For standard documents, pixels below T are assigned as black foreground, and those above become white background. For images with light strokes on a dark background (like `blackboard.png`), this logical condition is strictly inverted. The threshold T was manually tuned to find the optimal visual balance between text legibility and background noise reduction, resulting in the updated configurations: `receipt.png` (130), `blackboard.png` (55), `lilavati.png` (105), and `qr.png` (115).

Observations and Tuning Analysis

Tuning the global threshold manually highlighted the inherent difficulty of segmenting images with uneven lighting or localized degradations.

- **Receipt** ($T = 130$): The uniform lighting and high contrast between the dark ink and bright white paper made this image relatively easy to segment. A threshold of 130 perfectly isolated the text without pulling in the soft shadow on the bottom and right edges.
- **Lilavati Manuscript** ($T = 105$): The historical document contains varied ink densities and dark water stains. The updated threshold of 105 is stricter, dropping some of the lighter background stains, but faint ink strokes also begin to break apart, demonstrating the compromise required by global thresholds.
- **Blackboard** ($T = 55$, **inverted**): The illumination gradient and smeared chalk dust presented a challenge. A threshold of 55 (where pixels > 55 become text) captures the very faint chalk strokes on the left but introduces heavy chalk-dust noise artifacts on the brighter right side.
- **QR Code** ($T = 115$): This matrix exemplifies the failure state of global thresholding. Due to a severe diagonal lighting gradient, $T = 115$ captures the bright top-left region perfectly but completely obliterates the shadowed bottom-right region, turning it into a massive solid black block.

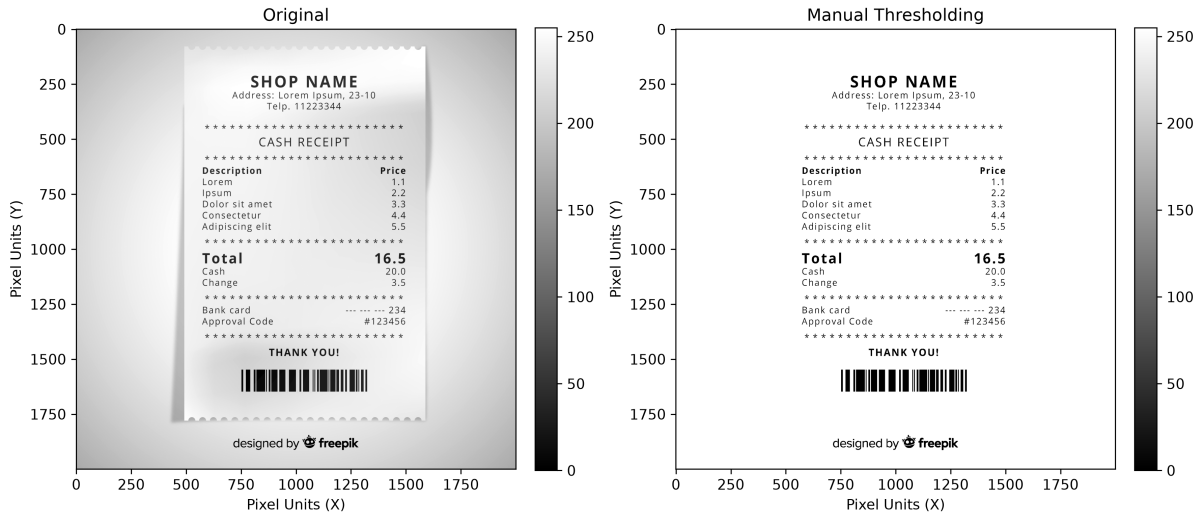


Figure 1: Manual Thresholding of receipt.png ($T = 130$).

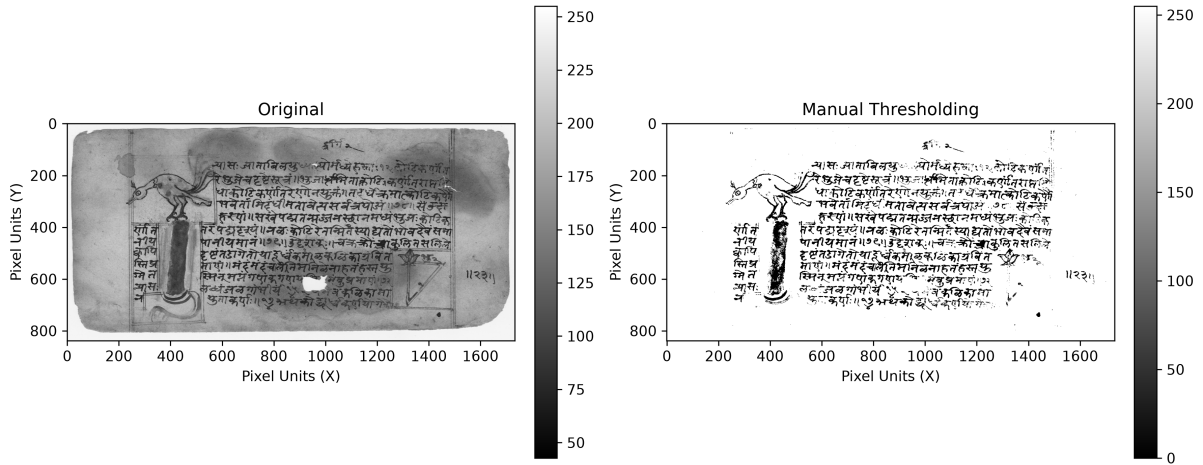


Figure 2: Manual Thresholding of lilavati.png ($T = 105$).

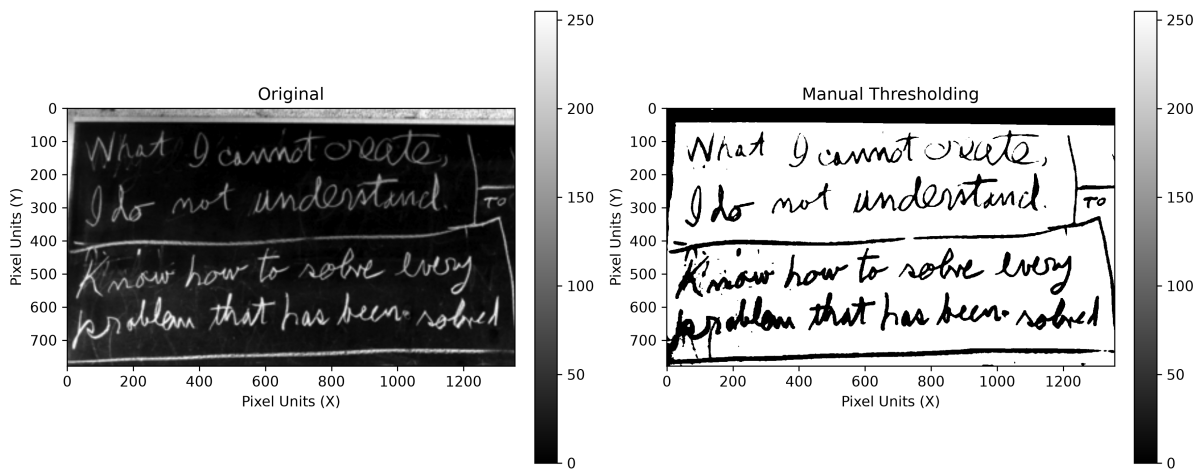


Figure 3: Manual Thresholding of blackboard.png ($T = 55$, inverted).

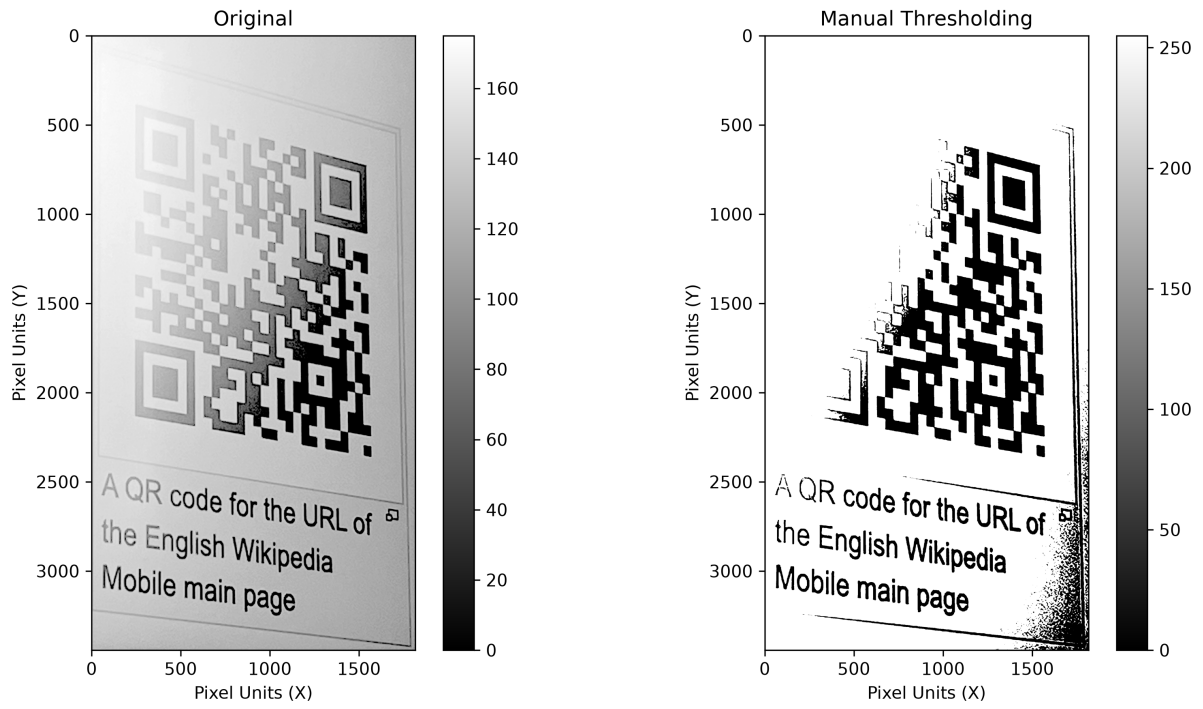


Figure 4: Manual Thresholding of `qr.png` ($T = 115$).